

Phase 4 - Project Planning

PocketSmart AI: Your Smart Budget & Recommendation Assistant

Date: 29 September 2026

1. Planning Overview

Project planning converts requirements and design into an ordered workflow. The project source defines five main milestones: Gemini initialization, core functionality, FastAPI integration, UI development, and testing/optimization.

2. Milestones

Milestone	Focus
1	Gemini AI Initialization
2	Core Functionalities Development
3	Backend - FastAPI Integration
4	UI Development
5	Testing & Optimization

3. Planning Objective

Complete the system progressively so that AI access is validated before recommendation logic, backend routes, frontend pages and final testing are integrated.

Project Planning - Milestones 1 and 2

4. Milestone 1 - Gemini AI Initialization

- Set up Google Cloud/platform access for Gemini API.
- Configure API keys, permissions and usage limits.
- Validate Gemini connectivity with a sample Python prompt.
- Test text-only and image + text prompts.
- Refine prompts for budget interpretation and recommendation generation.

5. Milestone 2 - Core Functionality

- Initialize FastAPI and required libraries.
- Load environment configuration.
- Create Home, Party and Jewelry endpoints.
- Integrate Gemini prompts and templates.
- Support optional image input for Jewelry.
- Connect recommendation results with platform/product sources.

Project Planning - Milestones 3 and 4

6. Milestone 3 - FastAPI Integration

- Implement /generate-home, /generate-party and /generate-jewelry.
- Implement /register, /login, /logout and /token.
- Implement /session-info and /session-data.
- Implement /recommendations-details and /history.
- Configure CORS and static routing.
- Implement startup configuration and main entry point.

7. Milestone 4 - UI Development

- Build responsive HTML forms.
- Create Home, Party and Jewelry planner pages.
- Create Register and Login pages.
- Create Dashboard and History pages.
- Add dynamic recommendation display using JavaScript.
- Use Jinja2 templates with FastAPI.

Project Planning - Testing, Risks and Outcome

8. Milestone 5 - Testing & Optimization

- Test real-world use cases across all three planners.
- Evaluate Gemini response quality and budget adherence.
- Check platform/source accuracy.
- Optimize prompts and validations.
- Improve session handling and UI/UX.
- Add fallback/default recommendations when AI results are insufficient.

9. Risk Management

Risk	Planned Handling
Gemini unavailable	Error handling and fallback/default recommendations
Invalid input	Validation before processing
Third-party data limitations	Available APIs or simulated/mock sourcing
Authentication problems	Test registration, login, token, session and logout
Image failure	Optional image input and validation

10. Planning Outcome

The planning phase provides a clear path from AI setup to backend, UI, testing, documentation and final demonstration.